IMPORT DATA USING TRANSFORM MAPS (SPREADSHEET)

Description:

This project focuses on importing structured employee data from an external spreadsheet into the ServiceNow platform using Import Sets and Transform Maps. The project simulates a real-world scenario where employee data is received in Excel format and needs to be imported into ServiceNow efficiently and accurately.

The spreadsheet data is first loaded into an Import Set table, which acts as a staging area. A Transform Map is then configured to map the source fields to the corresponding fields in the target table. After transformation, the employee records are created or updated in the target table.

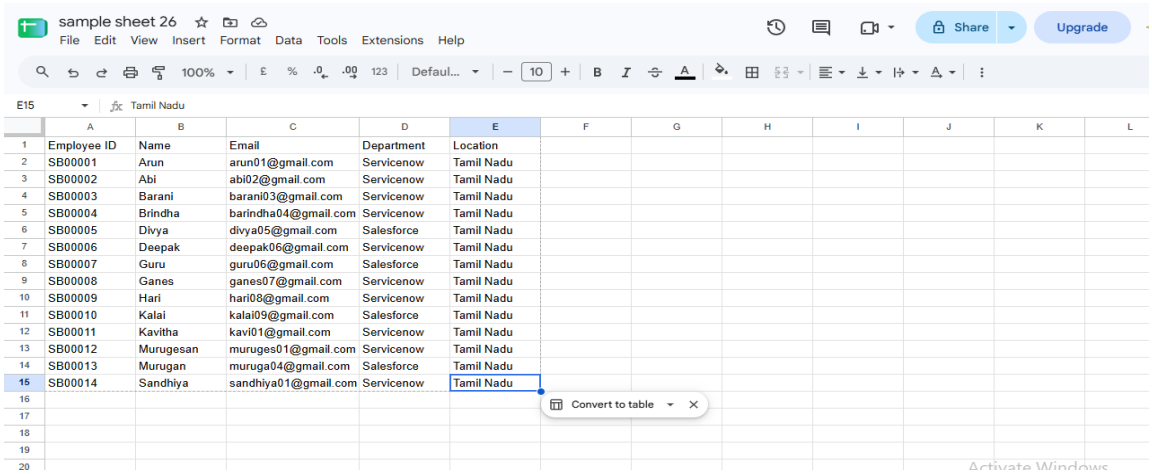
The project also uses Coalesce to prevent duplicate records and update existing employee information. Finally, Reports and Dashboards are created to analyse and monitor employee data.

PHASE 1: Prepare the Spreadsheet

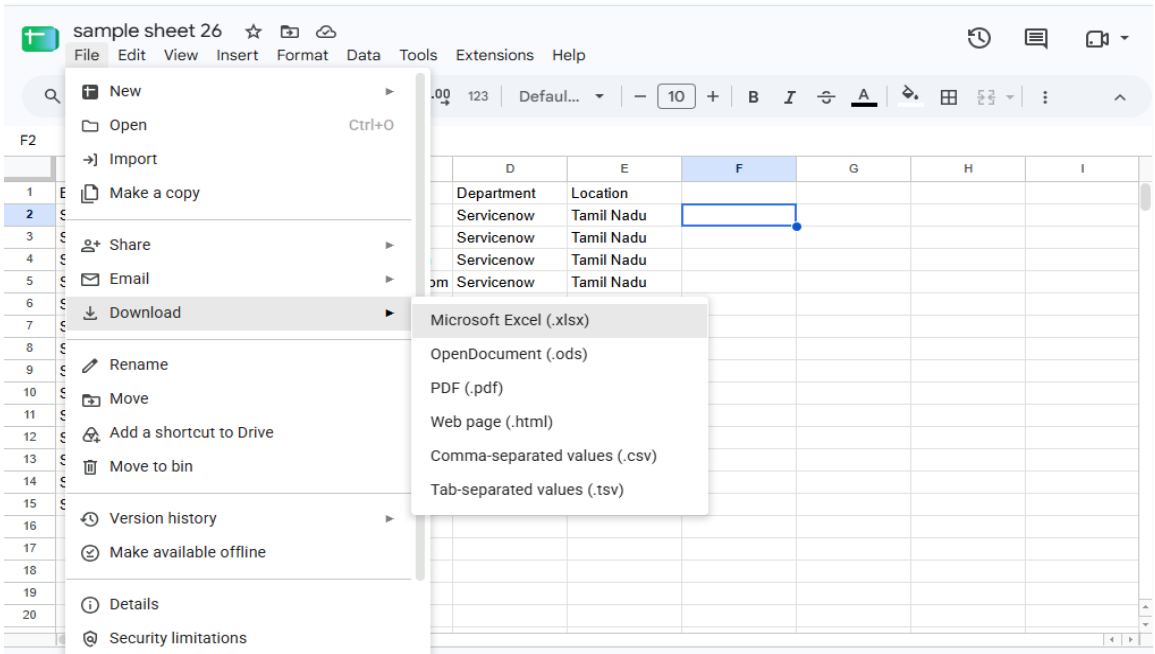
Steps:

1. Open Google Spreadsheet.
2. Create a new spreadsheet.
3. Enter the employee data with the required fields.
4. The spreadsheet contains the following fields:
    - Employee ID
    - Employee Name
    - Email
    - Department
    - Location
5. Enter sample employee records in the spreadsheet.
6. Download the spreadsheet in **Microsoft Excel (.xlsx)** format.
7. Save the file as **Sample Spreadsheet.xlsx**.

Sample Spreadsheet:



Employee ID	Name	Email	Department	Location
SB00001	Arun	arun01@gmail.com	Servicenow	Tamil Nadu
SB00002	Abi	abi02@gmail.com	Servicenow	Tamil Nadu
SB00003	Barani	barani03@gmail.com	Servicenow	Tamil Nadu
SB00004	Brindha	barindha04@gmail.com	Servicenow	Tamil Nadu
SB00005	Divya	divya05@gmail.com	Salesforce	Tamil Nadu
SB00006	Deepak	deepak06@gmail.com	Servicenow	Tamil Nadu
SB00007	Guru	guru06@gmail.com	Salesforce	Tamil Nadu
SB00008	Ganes	ganes07@gmail.com	Servicenow	Tamil Nadu
SB00009	Hari	hari08@gmail.com	Servicenow	Tamil Nadu
SB00010	Kalai	kalai09@gmail.com	Salesforce	Tamil Nadu
SB00011	Kavitha	kavi01@gmail.com	Servicenow	Tamil Nadu
SB00012	Murugesan	muruges01@gmail.com	Servicenow	Tamil Nadu
SB00013	Murugan	muruga04@gmail.com	Salesforce	Tamil Nadu
SB00014	Sandhiya	sandhiya01@gmail.com	Servicenow	Tamil Nadu



PHASE 2: Create New Custom Table

In this phase, a custom table is created in ServiceNow to store the employee information imported from the spreadsheet.

Navigation:

Application Navigator → Tables → Create New

Table Details:

Label: Employee Test

Name: `u_employee_test`

After creating and saving the table, open the table form.

Create Employee ID Field

1. Click **Show Form**.
2. Open **Form Context Menu / Additional Actions**.
3. Select **Configure → Form Layout**.
4. Click **New** to create a new field.
5. Enter the following details:

   Column Label: Employee ID
   Type: String

6. Click **Add**.

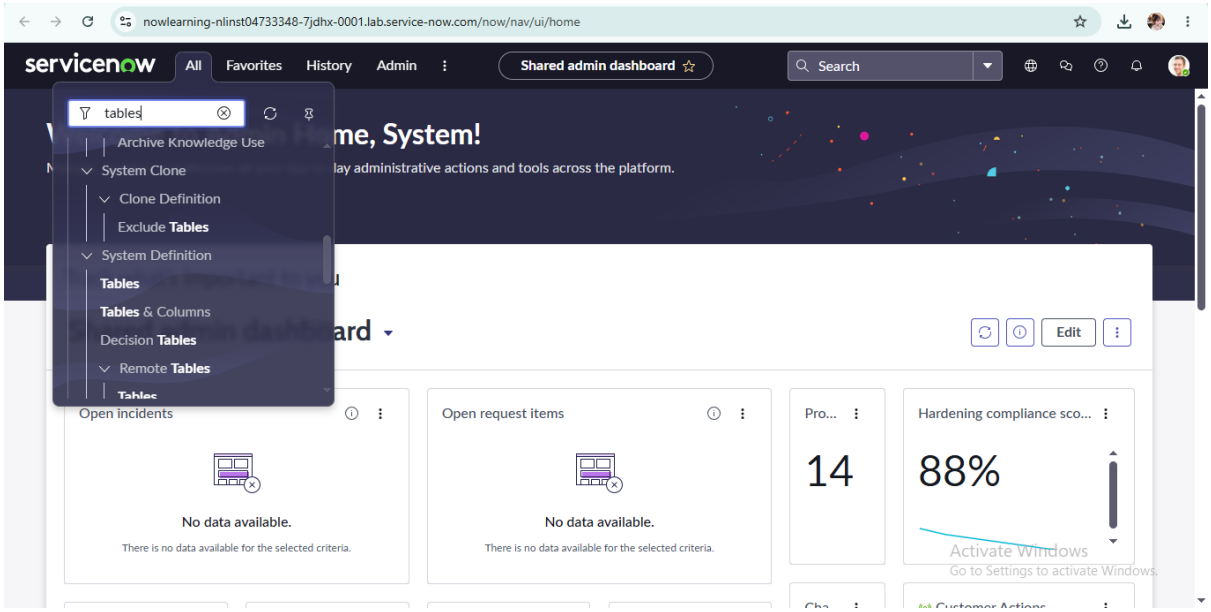
Create Remaining Fields

Create the following fields using the same procedure:

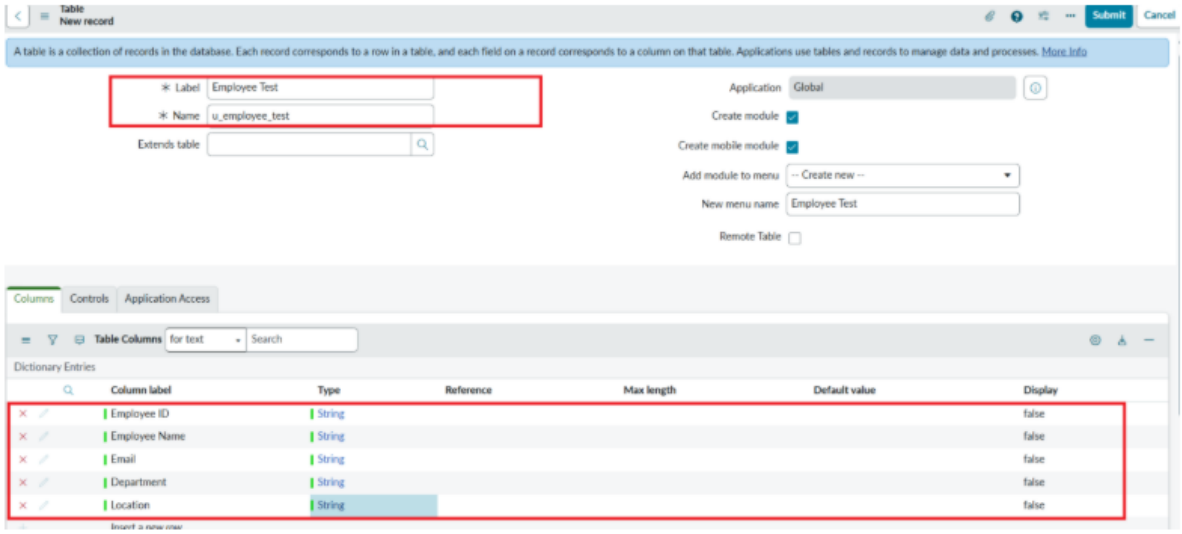
Field	Type
Employee ID	String
Employee Name	String
Email	String
Department	String
Location	String

7. Click **Save**.
8. Verify that all fields are displayed in the Employee Test table.

Employee Test Table



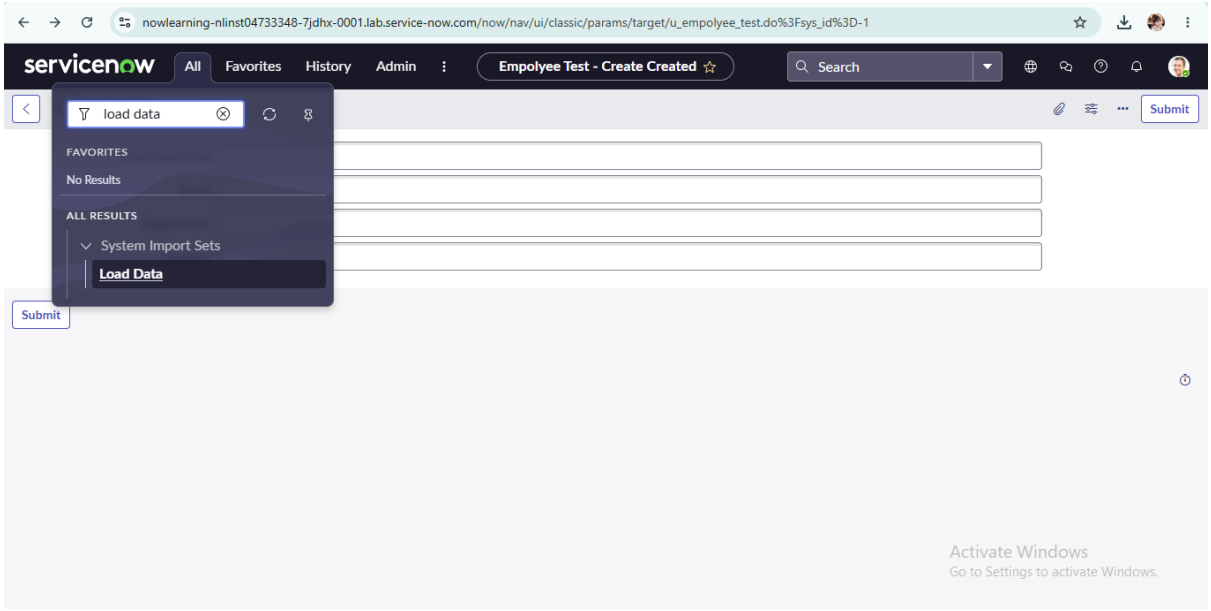
Employee Test Fields



Phase 3: Import Set Table

Navigation:

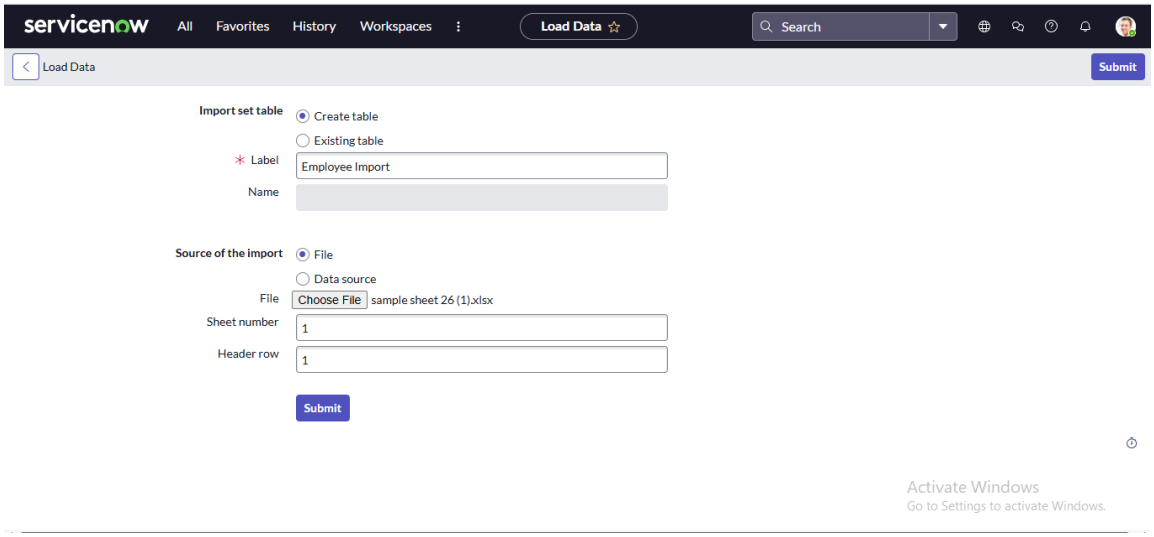
In **Application Navigator → Load Data**



Click on **Load Data** to create an Import Set Table.

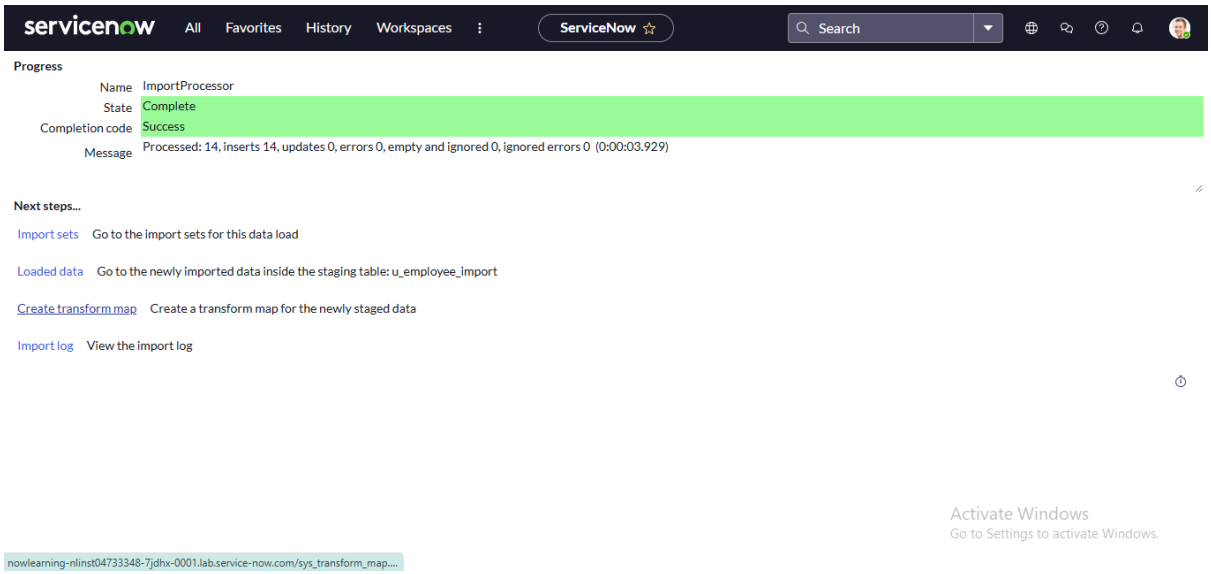
Fill in the details as shown below:

- **Label:** Employee Import
- **Name:** `u_employee_import` *(auto populated)*



Click on **Submit**.

Once you click **Submit**, a success message will be displayed.



Phase 4: Create Transform Map

Navigation:

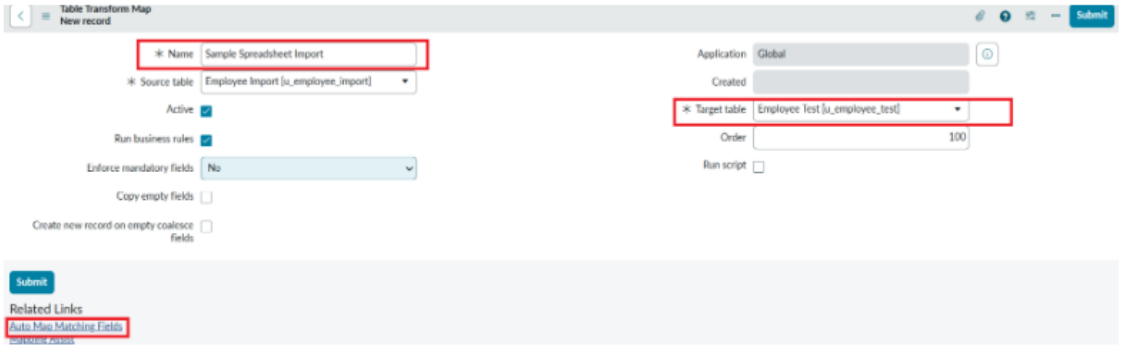
After creating the **Import Set Table**, click on **Create Transform Map**.

A Transform Map is used to map the fields from the **Source Table (Employee Import)** to the **Target Table (Employee Test)**.

Steps:

1. Click on **Create Transform Map**.
2. Fill in the details as shown below:

- **Name:** Sample Spreadsheet Import
- **Source Table:** Employee Import
- **Target Table:** Employee Test



3. Click on **Auto Map Matching Fields**.
4. The matching fields will be mapped automatically between the source and target tables.
5. If required, click on **Mapping Assist** to manually map the fields.

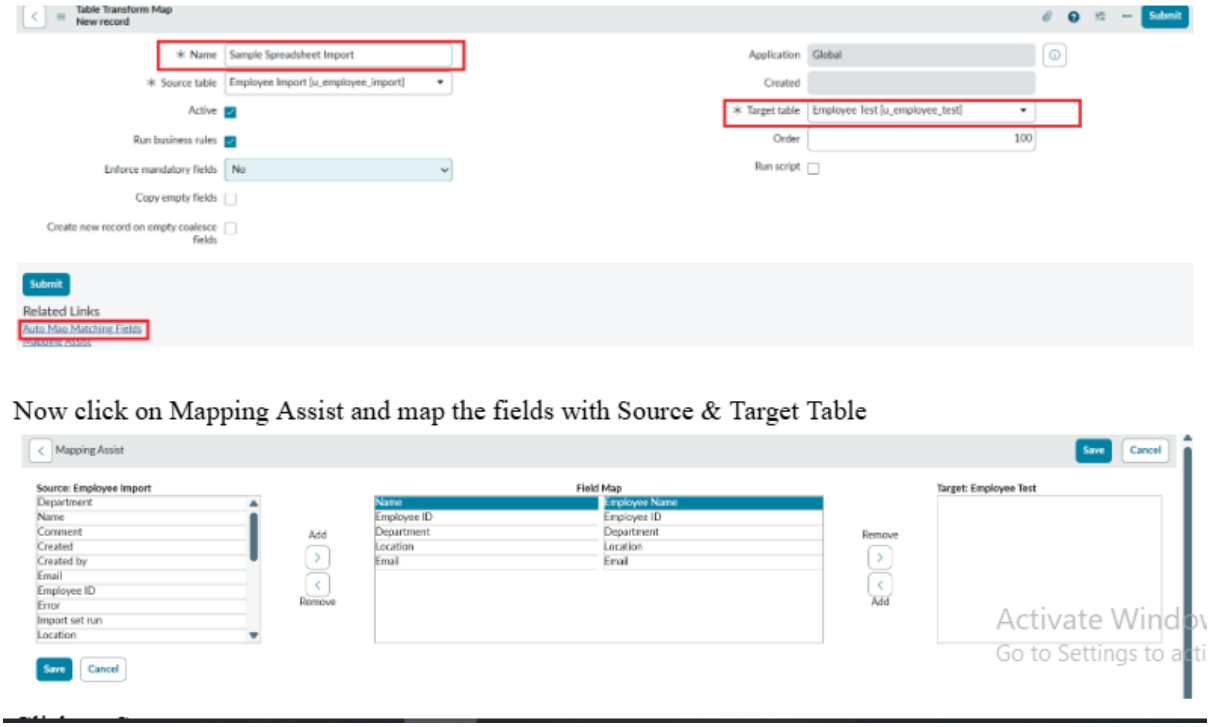
Field Mapping:

Source Table – Employee Import	**Target Table – Employee Test**
Employee ID	Employee ID

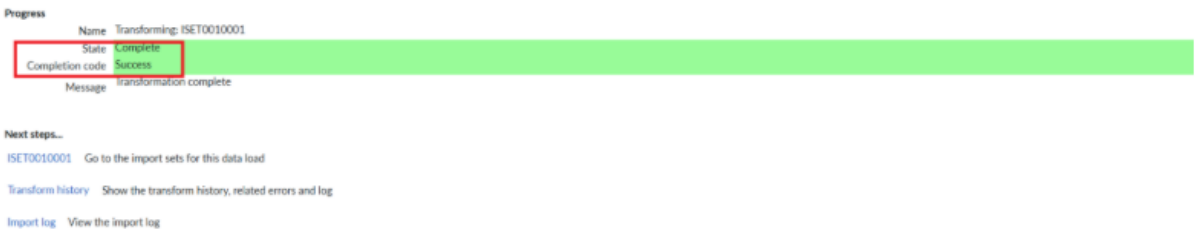
Source Table – Employee Import Target Table – Employee Test

Employee Name	Employee Name
Email	Email
Department	Department
Location	Location

6. Verify that all the required source and target fields are mapped correctly.
7. Click on **Save**.



8. Make sure that both the **Source Table** and **Target Table** fields are mapped correctly as shown below.

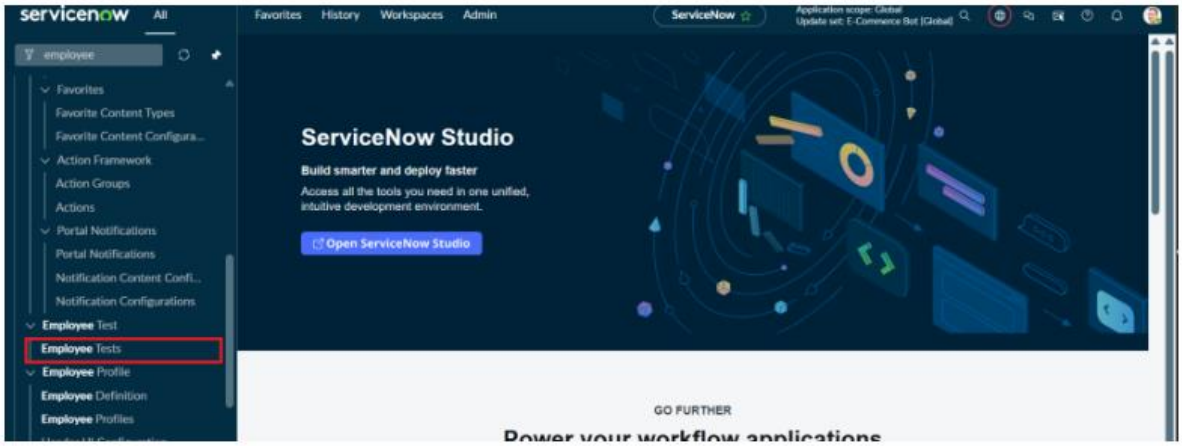


Phase 5: Transform Data & Validate

Navigation:

Open the created **Transform Map**.

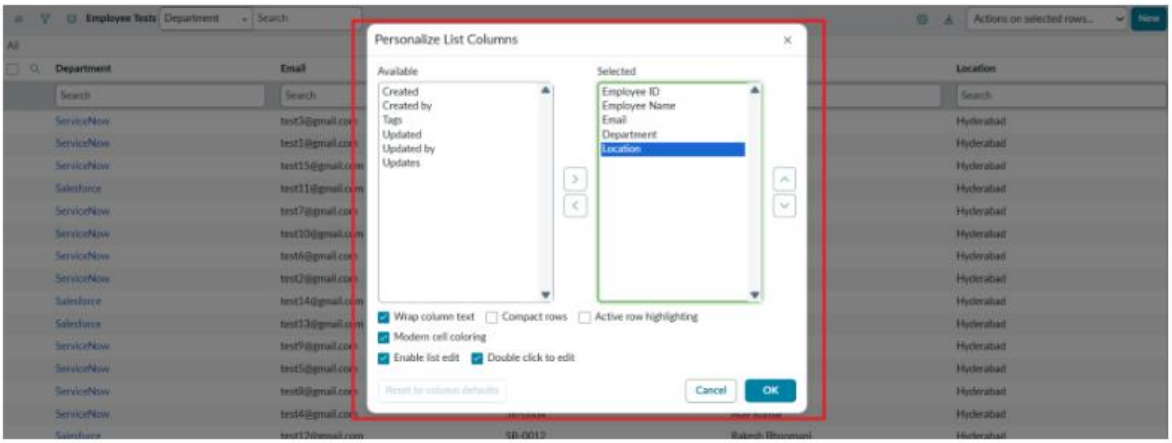
Click on **Transform** to start importing the data.



Click on the **Transform** button to execute the transformation.

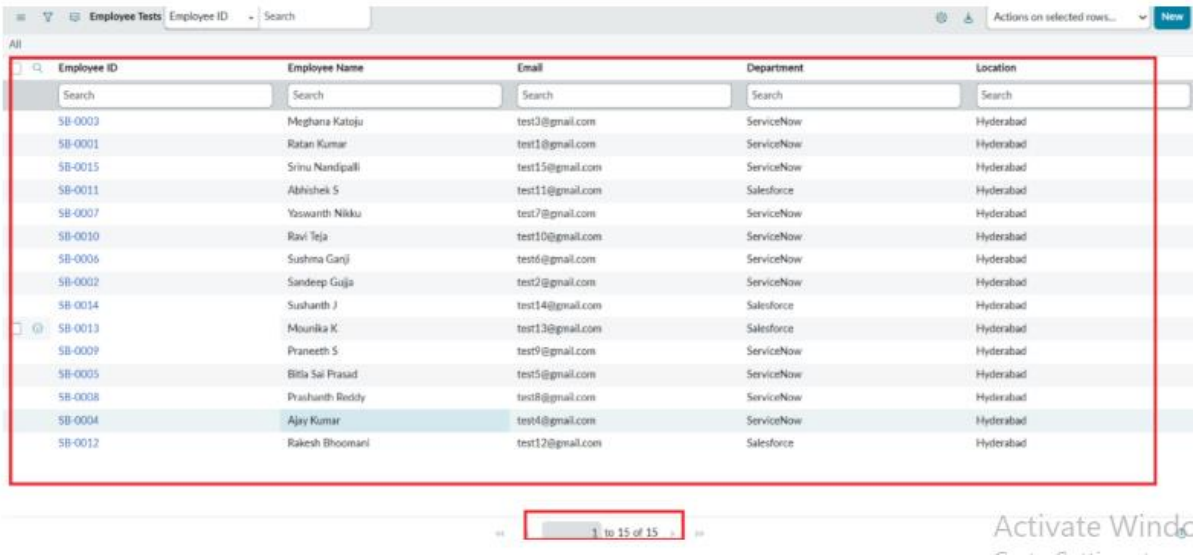


Once the transformation is completed, verify that the **success message** is displayed.



Next, open the **Employee Test** table from the Application Navigator.

Verify that the employee records from the spreadsheet have been successfully imported into the target table.



If required, use **Personalize List Columns** to arrange the fields and click **OK**.

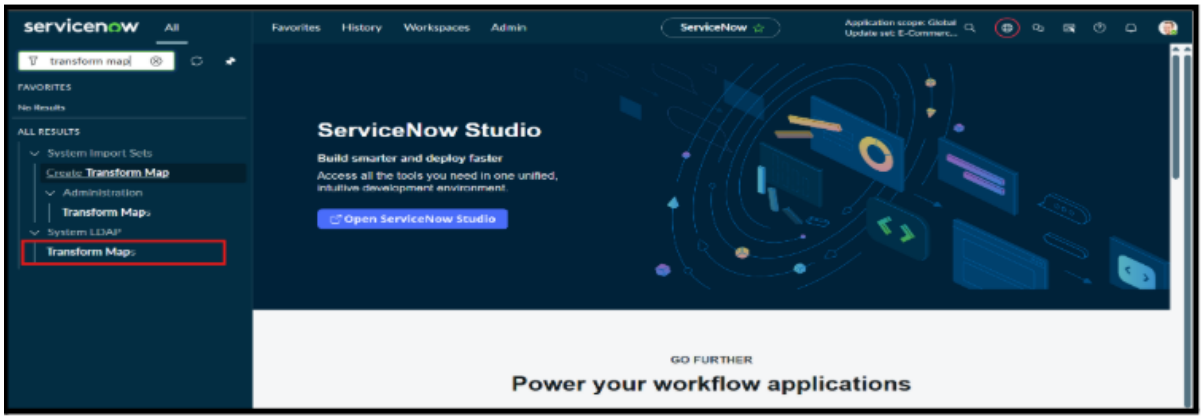
Final Result:

All employee records are successfully imported into the **Employee Test** table using the Import Set and Transform Map.

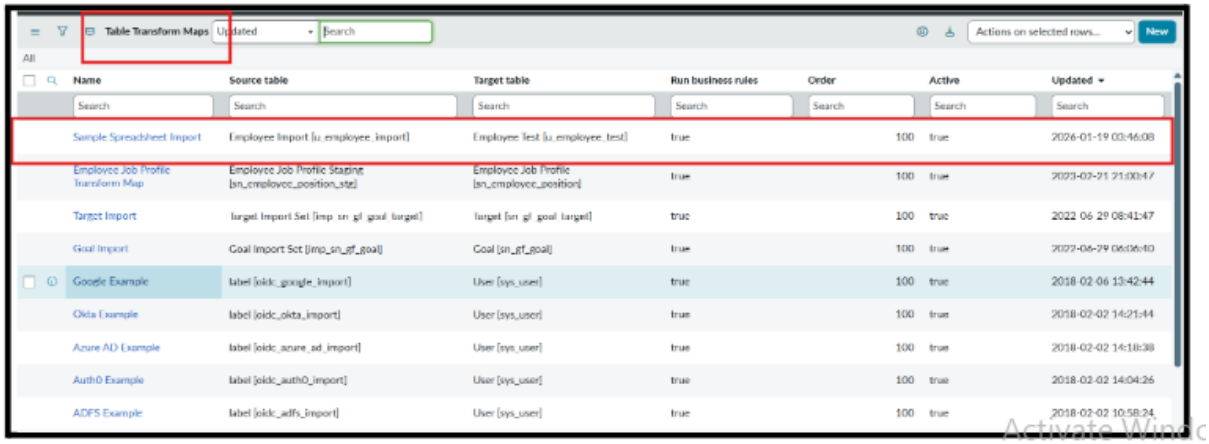
Phase 6: Enable Coalesce to Avoid Duplicate Records

Navigation:

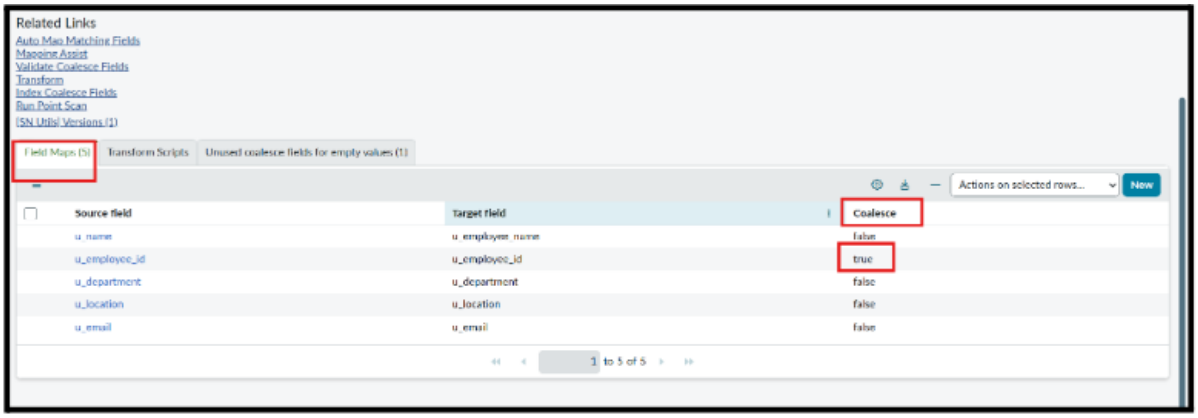
Open the created **Transform Map – Sample Spreadsheet Import**.



Go to the **Field Maps** section.



Find the **Employee ID** field and enable **Coalesce** by setting it to **True**.



Click on **Save**.

Purpose of Coalesce:

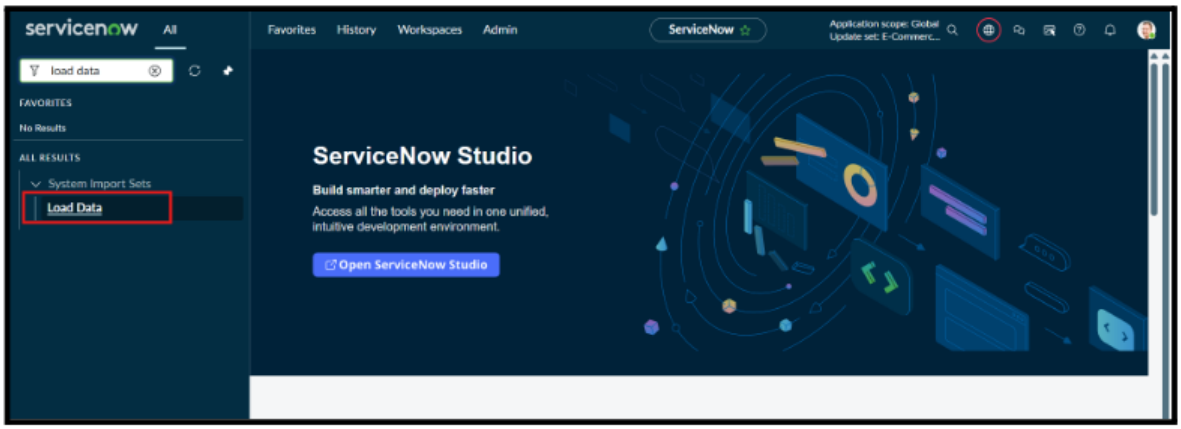
Coalesce helps to identify existing records using the **Employee ID** field. If the Employee ID already exists, the existing record can be updated instead of creating a duplicate record.

PHASE 7: Inserting New Data in Excel Format

After enabling **Coalesce**, add new employee records to the Excel spreadsheet.

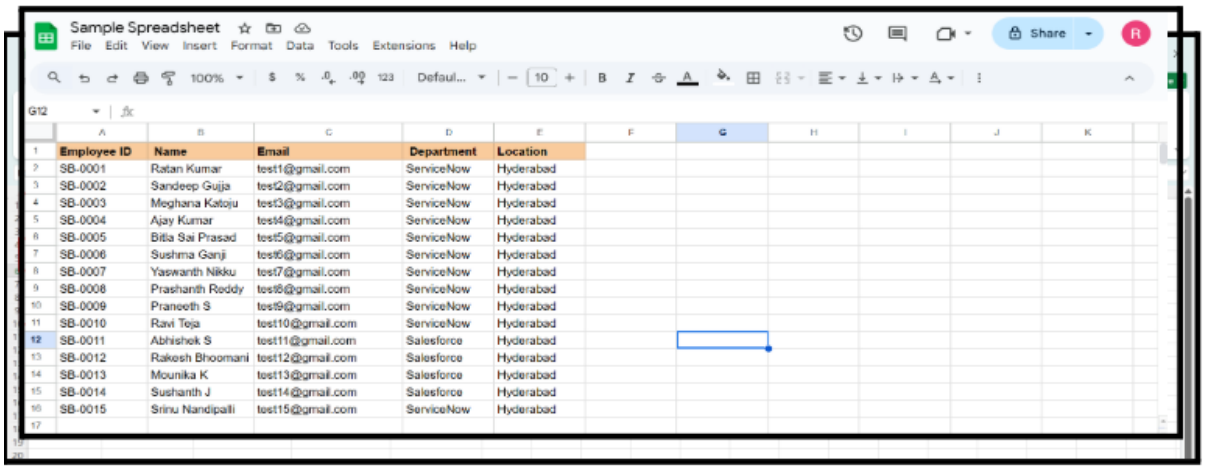
Steps:

1. Open the Excel spreadsheet used for importing employee data.
2. Add a **new employee record** with a new **Employee ID**.
3. Save the Excel file.
4. Go to **Load Data** in ServiceNow.
5. Upload the updated Excel file.
6. Run the **Transform** process using the existing Transform Map.
7. Open the **Employee Test** table and verify the new employee record.



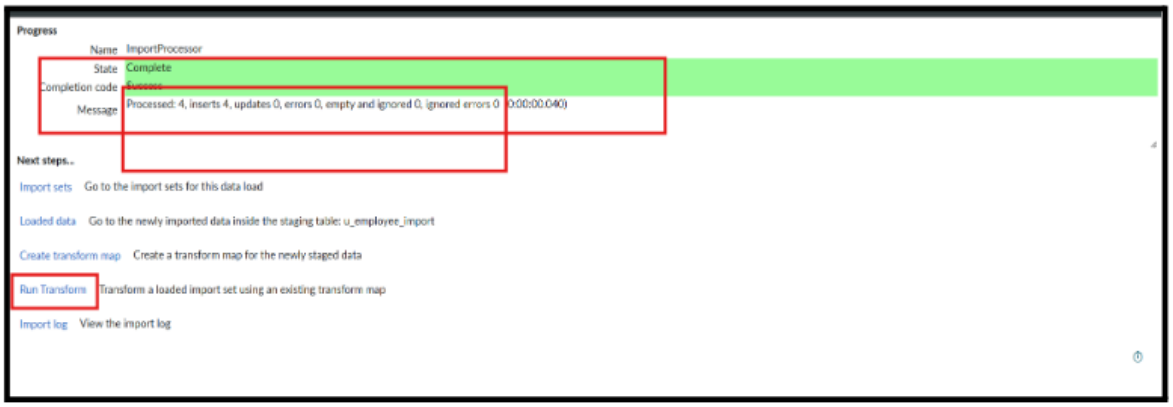
Testing Coalesce:

- If a **new Employee ID** is added, a **new record** is inserted.



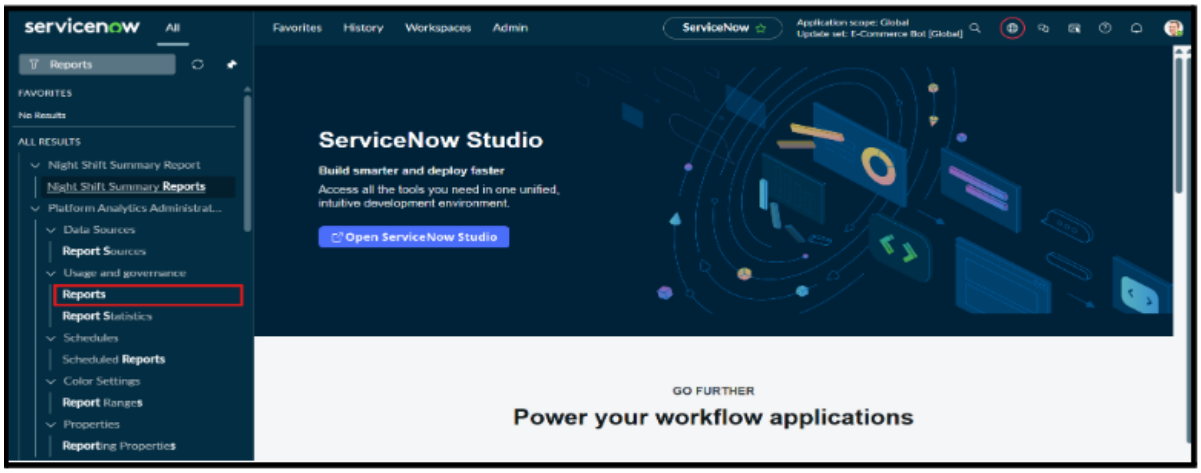
Employee ID	Name	Email	Department	Location
SB-0001	Ratan Kumar	test1@gmail.com	ServiceNow	Hyderabad
SB-0002	Sandeep Gujja	test2@gmail.com	ServiceNow	Hyderabad
SB-0003	Meghana Katoju	test3@gmail.com	ServiceNow	Hyderabad
SB-0004	Ajay Kumar	test4@gmail.com	ServiceNow	Hyderabad
SB-0005	Bitla Sai Prasad	test5@gmail.com	ServiceNow	Hyderabad
SB-0006	Sushma Ganji	test6@gmail.com	ServiceNow	Hyderabad
SB-0007	Yaswanth Nikku	test7@gmail.com	ServiceNow	Hyderabad
SB-0008	Prashanth Reddy	test8@gmail.com	ServiceNow	Hyderabad
SB-0009	Praneeth S	test9@gmail.com	ServiceNow	Hyderabad
SB-0010	Ravi Teja	test10@gmail.com	ServiceNow	Hyderabad
SB-0011	Abhishek S	test11@gmail.com	Salesforce	Hyderabad
SB-0012	Rakesh Bhoomani	test12@gmail.com	Salesforce	Hyderabad
SB-0013	Mounika K	test13@gmail.com	Salesforce	Hyderabad
SB-0014	Sushanth J	test14@gmail.com	Salesforce	Hyderabad
SB-0015	Srinu Nandipalli	test15@gmail.com	ServiceNow	Hyderabad

- If an **existing Employee ID** is added with changed information, the existing record is **updated**.
- If the same data is imported again without changes, it is **ignored** rather than creating a duplicate.



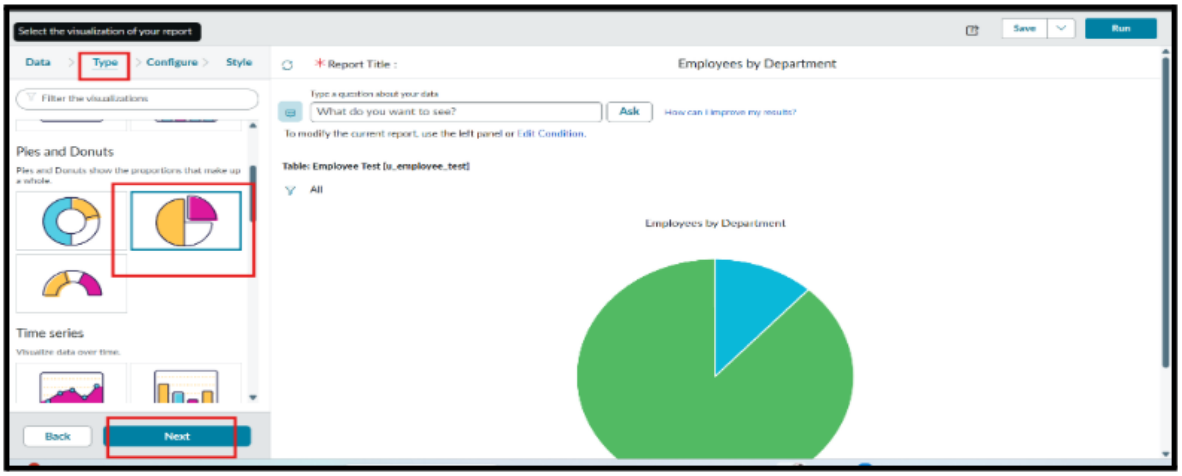
PHASE 8: Create Reports on Employee Test Table

Create reports using the employee records available in the **Employee Test** table.



Report 1: Employees by Department

- Open **Reports → Create New**
- Select the **Employee Test** table.
- Select **Pie Chart** as the report type.
- Set **Group by:** Department
- Set **Aggregation:** Count
- Enter the report name as **Employees by Department**.
- Click **Save**.



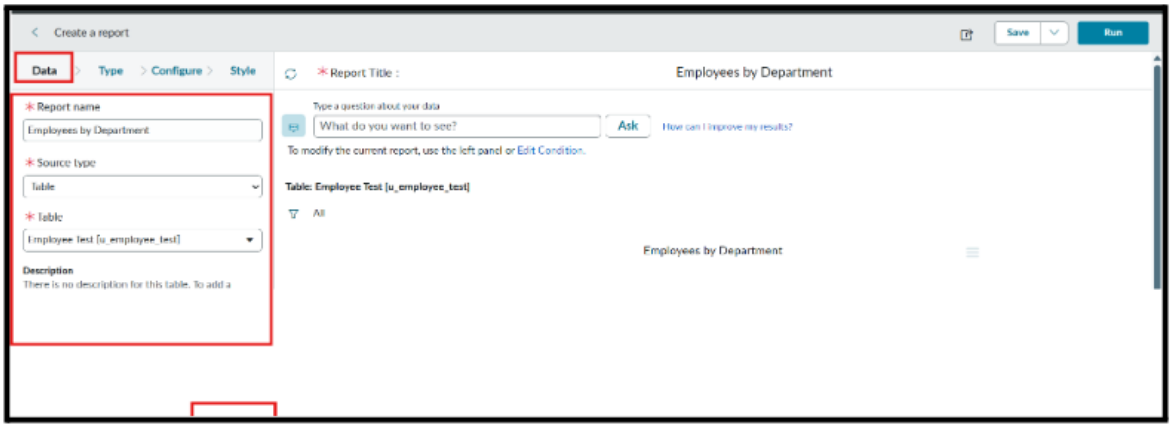
Report 2: Employees by Location

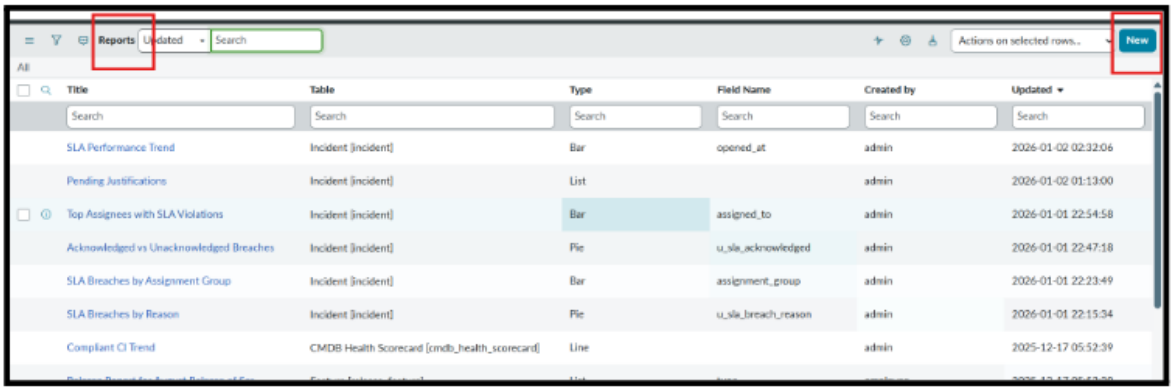
- Create another report using the **Employee Test** table.
- Select **Bar Chart** as the report type.
- Set **Group by:** Location
- Set **Aggregation:** Count
- Enter the report name as **Employees by Location**.
- Click **Save**.



Report 3: Employee List Report

- Create a new report using the **Employee Test** table.
- Select **List** as the report type.
- Add the following fields:
  - Employee ID
  - Employee Name
  - Email
  - Department
  - Location
- Enter the report name as **Employee List Report**.
- Click **Save**.





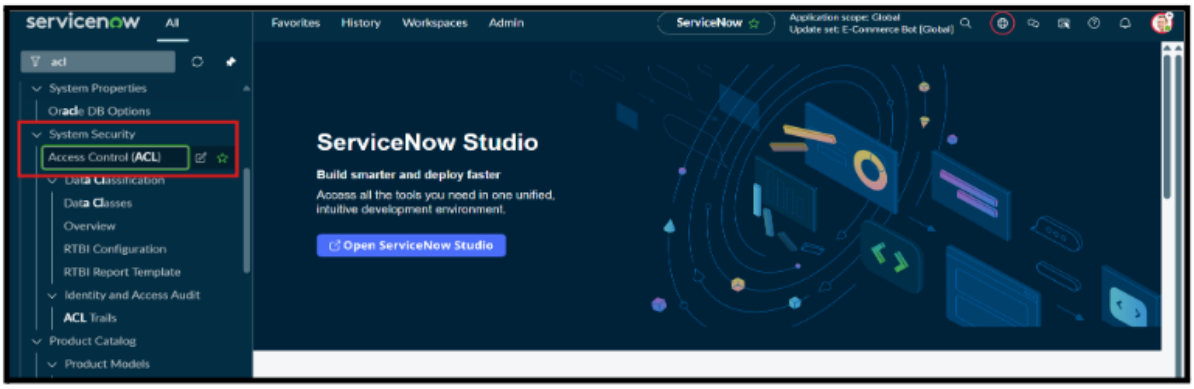
These reports help to view and analyse employee information based on **Department, Location, and employee details**.

PHASE 9: Add Reports to Dashboard

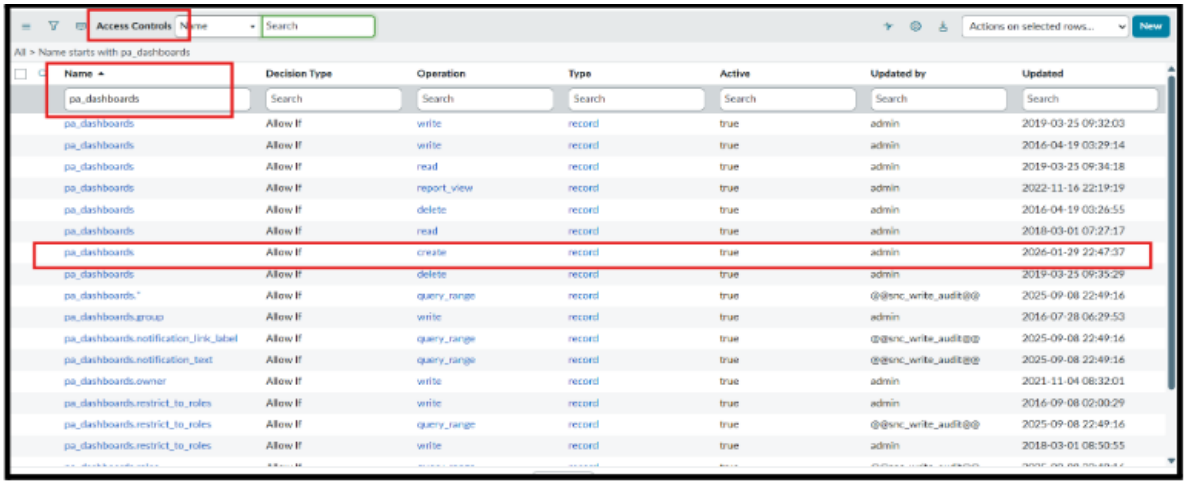
After creating the reports, add them to a dashboard to view the employee information in one place.

Steps:

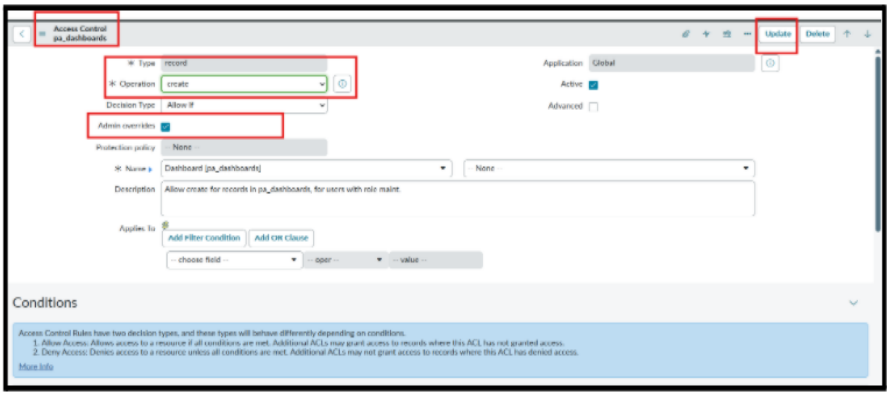
1. Open **Self-Service → Dashboards**.



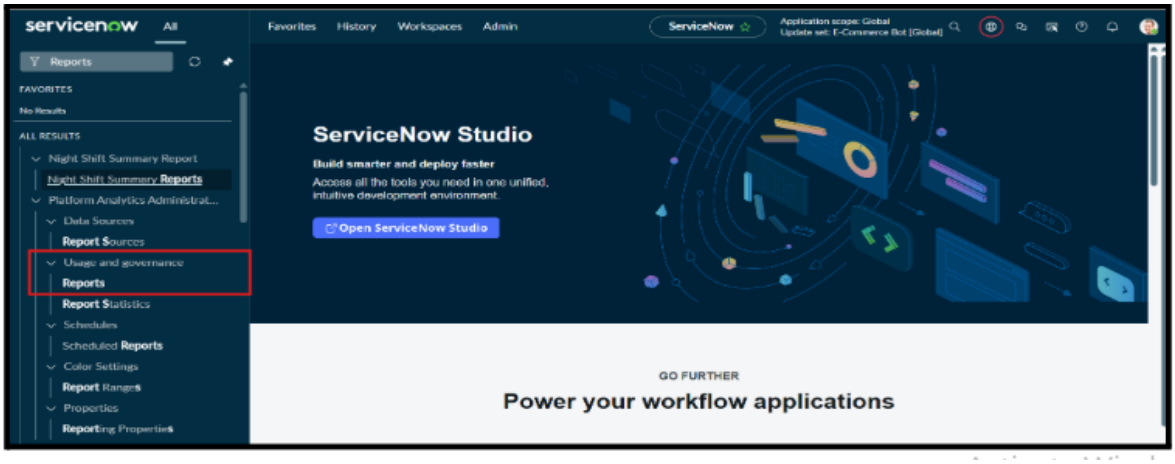
2. Create a new dashboard.
3. Enter the dashboard name as **Employee Analytics Dashboard**.

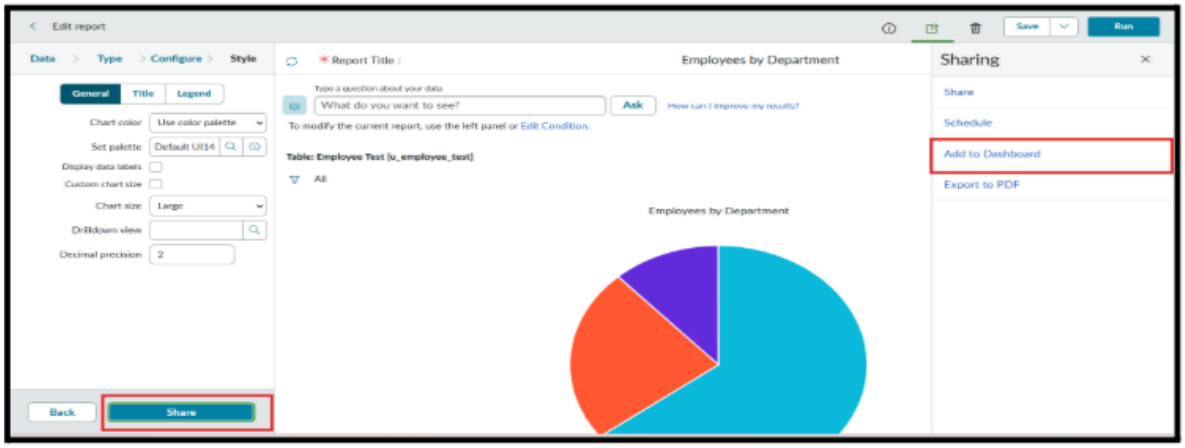


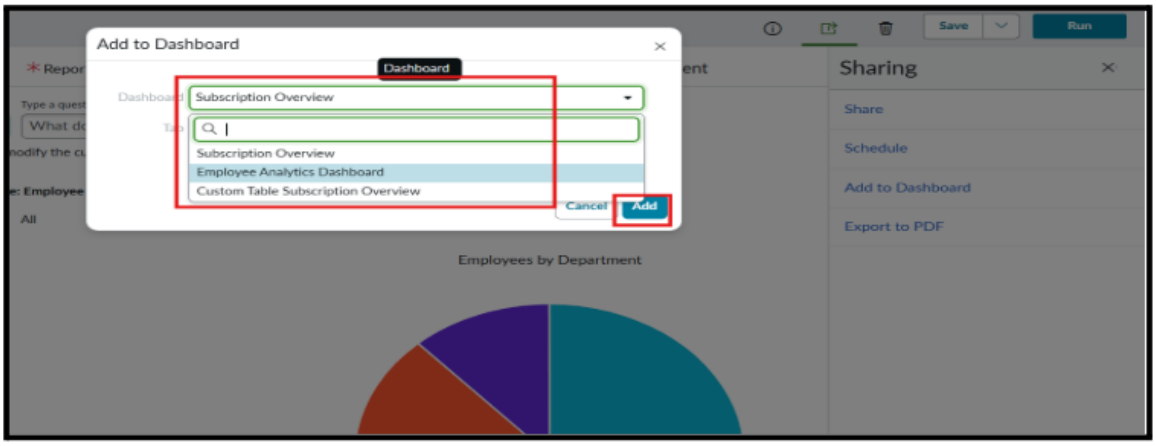
4. Open the created dashboard.
5. Click **Add Content**.
6. Add the following reports:
    - **Employees by Department**
    - **Employees by Location**
    - **Employee List Report**



7. Arrange the reports on the dashboard as required.
8. Save the dashboard.







Final Result:

The dashboard displays the employee reports together, making it easier to view and analyse the imported employee data.

Conclusion:

1. This project successfully implemented employee data management using **ServiceNow Import Sets and Transform Maps**, providing an efficient way to import and manage employee information.
2. **Coalesce** was enabled on the **Employee ID** field to identify existing records and prevent duplicate entries. This allows existing employee information to be updated during repeated imports.
3. **Reports and Dashboards** were created to analyse employee data, including department-wise and location-wise employee information and employee records.
4. These reports were added to an **Employee Analytics Dashboard**, providing a centralized view of employee data and making it easier to monitor and analyse the imported information.